

Comparative LLM Analysis: USA Vs China:

Team Members

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Business Objective:

- ▶ Conduct a comparative analysis of LLMs from USA and China
- ▶ **Models considered:** LLaMA, Qwen, and DistilBERT models
- ▶ We are Chosen (Question Answering) Task

Dataset: ▶ Question Answering:

- ▶ * SQuAD v2 - English, CMRC2018 - Chinese
- ▶ * We are done EDA:
- ▶ 1.SQuAD Cleaned Samples:

	Original Question	Cleaned Question	Original Answer
0	When did Beyonce start becoming popular?	when did beyonce start becoming popular?	in the late 1990s
1	What areas did Beyonce compete in when she was...	what areas did beyonce compete in when she was...	singing and dancing
2	When did Beyonce leave Destiny's Child and bec...	when did beyonce leave destiny's child and bec...	2003
3	In what city and state did Beyonce grow up?	in what city and state did beyonce grow up?	Houston, Texas
4	In which decade did Beyonce become famous?	in which decade did beyonce become famous?	late 1990s

2. CMRC Cleaned Samples:

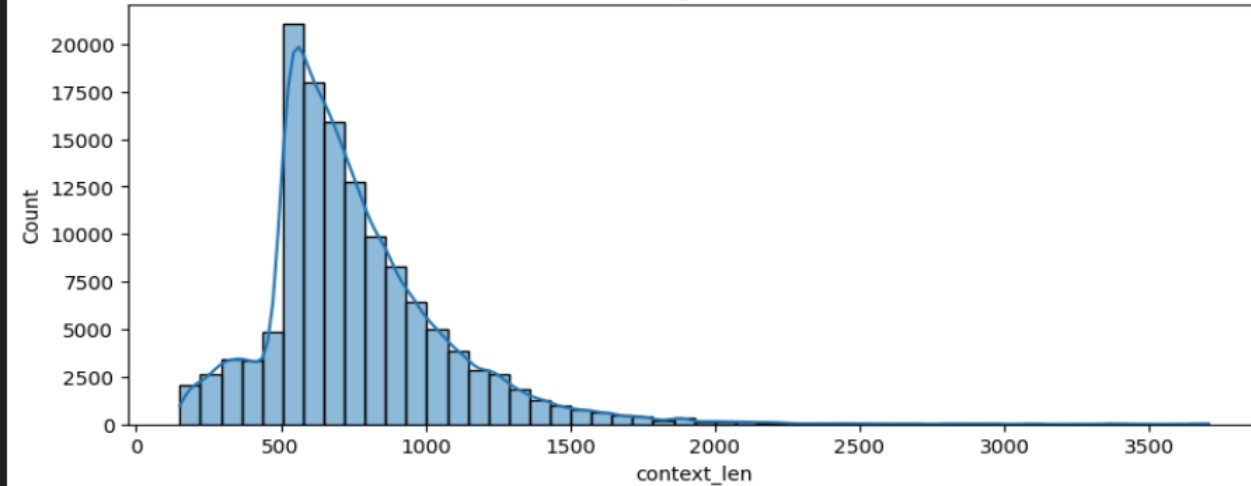
	Original Question	Cleaned Question	Original Answer
0	范廷颂是什么时候被任为主教的?	范廷颂是什么时候被任为主教的?	1963年
1	1990年, 范廷颂担任什么职务?	1990年, 范廷颂担任什么职务?	1990年被擢升为天主教河内总教区宗座署理
2	范廷颂是于何时何地出生的?	范廷颂是于何时何地出生的?	范廷颂于1919年6月15日在越南宁平省天主教发艳教区出生
3	1994年3月, 范廷颂担任什么职务?	1994年3月, 范廷颂担任什么职务?	1994年3月23日, 范廷颂被教宗若望保禄二世擢升为天主教河内总教区总主教并兼天主教谅山教区...
4	范廷颂是何时去世的?	范廷颂是何时去世的?	范廷颂于2009年2月22日清晨在河内离世

Data Analysis:

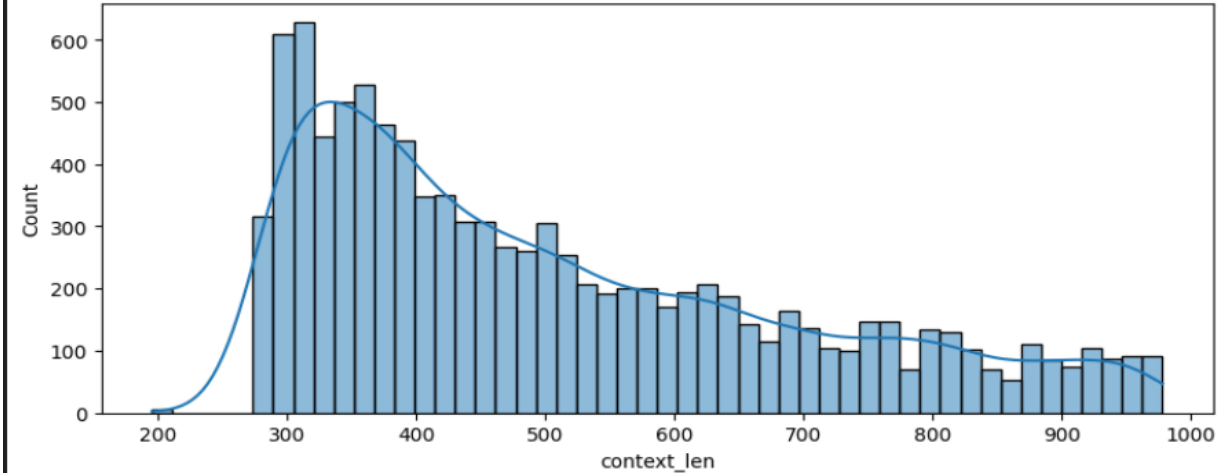
Plot distributions:

- context lengths and question lengths

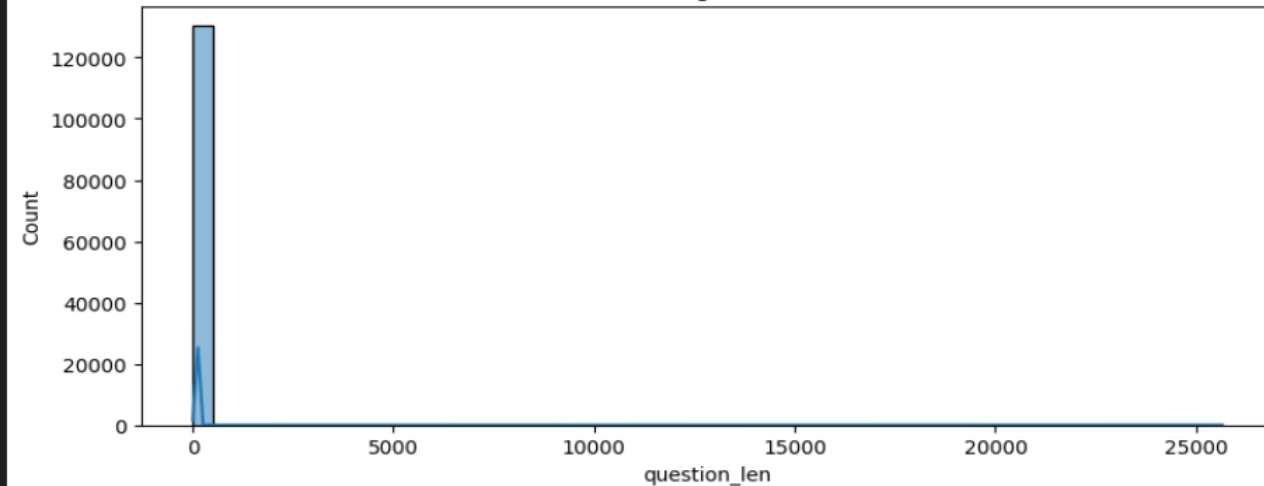
SQuAD: Context length distribution (chars)



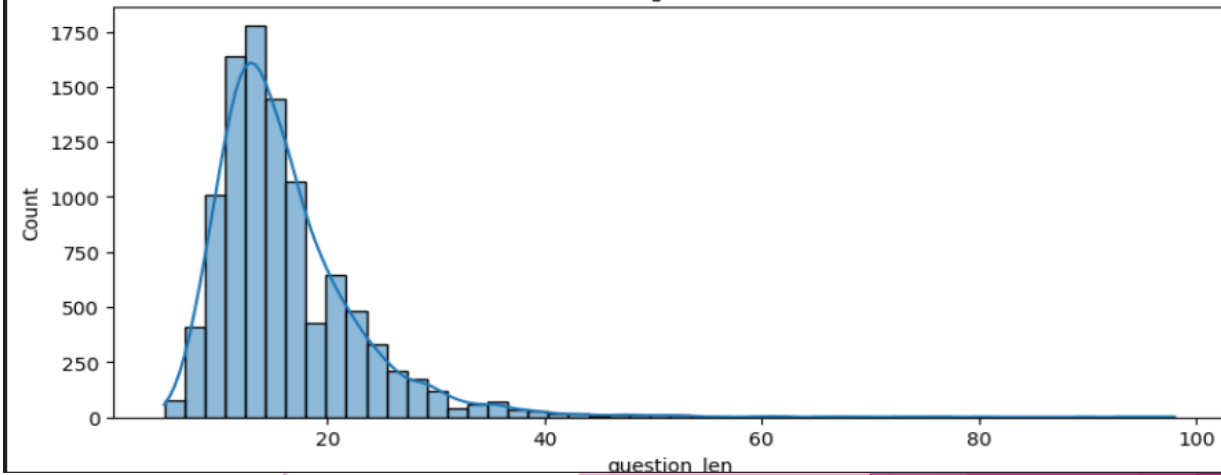
CMRC: Context length distribution (chars)



SQuAD: Question length distribution (chars)



CMRC: Question length distribution (chars)



QA Samples:

English QA Sample (SQuAD) vs Chinese QA Sample (CMRC):

English QA Sample (SQuAD):

Context: The Normans (Norman: Nourmands; French: Normands; Latin: Normanni) were the people who in the 10th a...

Question: In what country is Normandy located?

Gold Answer: France

Chinese QA Sample (CMRC):

Context: 《战国无双3》()是由光荣和ω-force开发的战国无双系列的正统第三续作。本作以三大故事为主轴,分别是以武田信玄等人为主的《关东三国志》,织田信长等人为主的《战国三杰》,石田三成等人为主的《关原的年

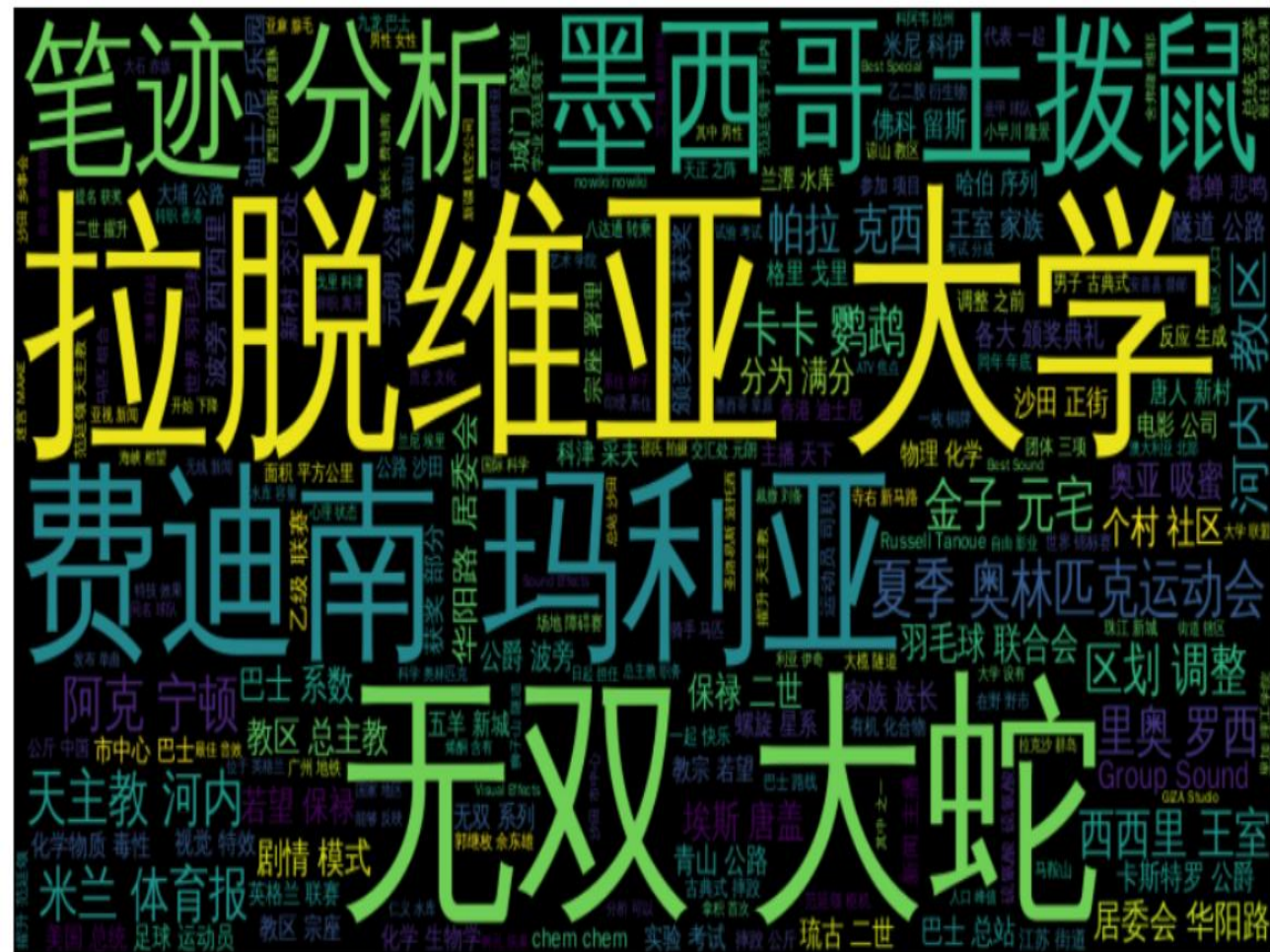
Question: 《战国无双3》是由哪两个公司合作开发的?

Gold Answer: 光荣和ω-force

Word Clouds:

* (Chinese) word cloud vs Word cloud (English):

Chinese word cloud (CMRC contexts)

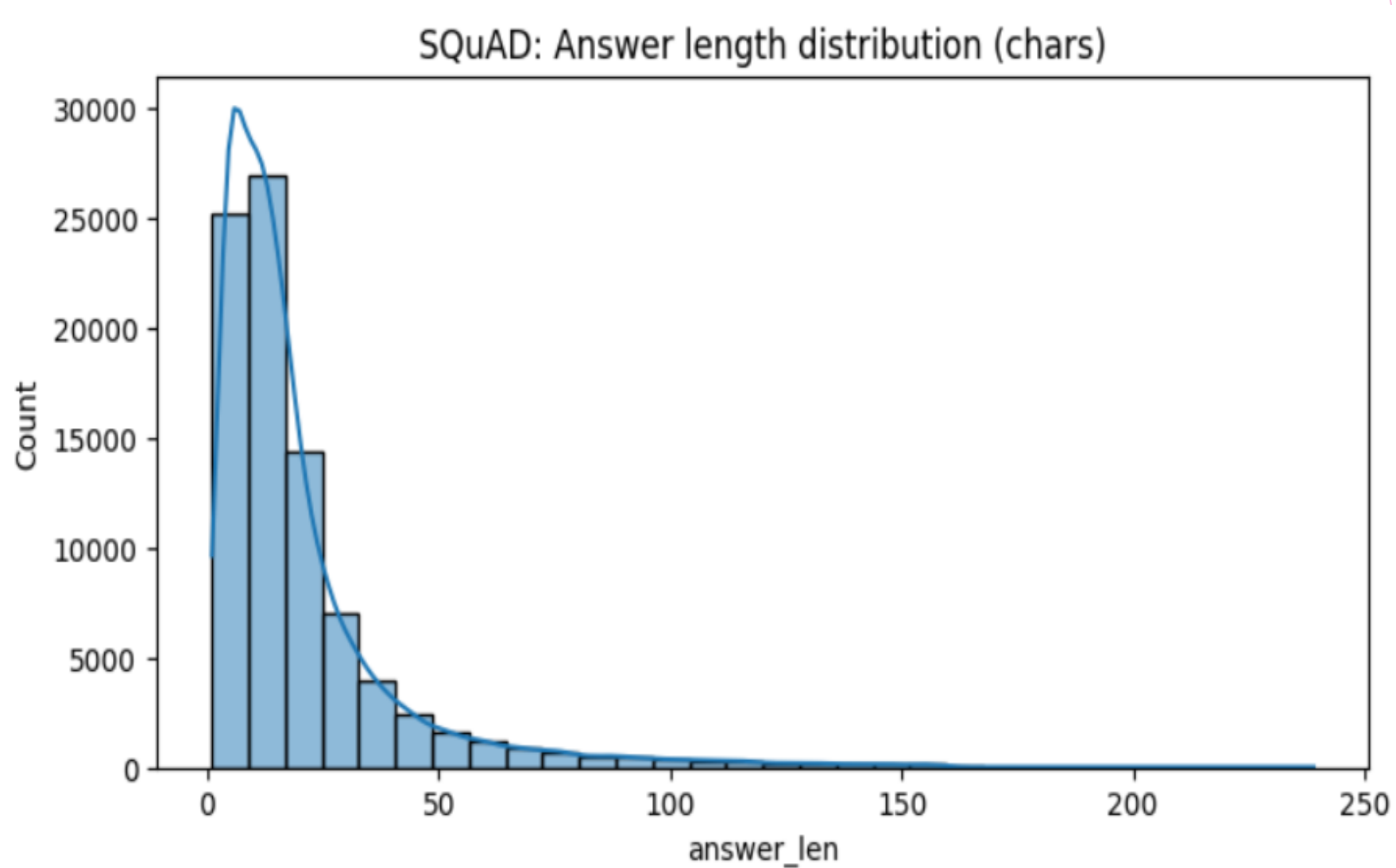


English word cloud (SQuAD contexts)



Answer Distribution:

* SQuAD answer presence and answer length distribution:



Evaluation Metrics:

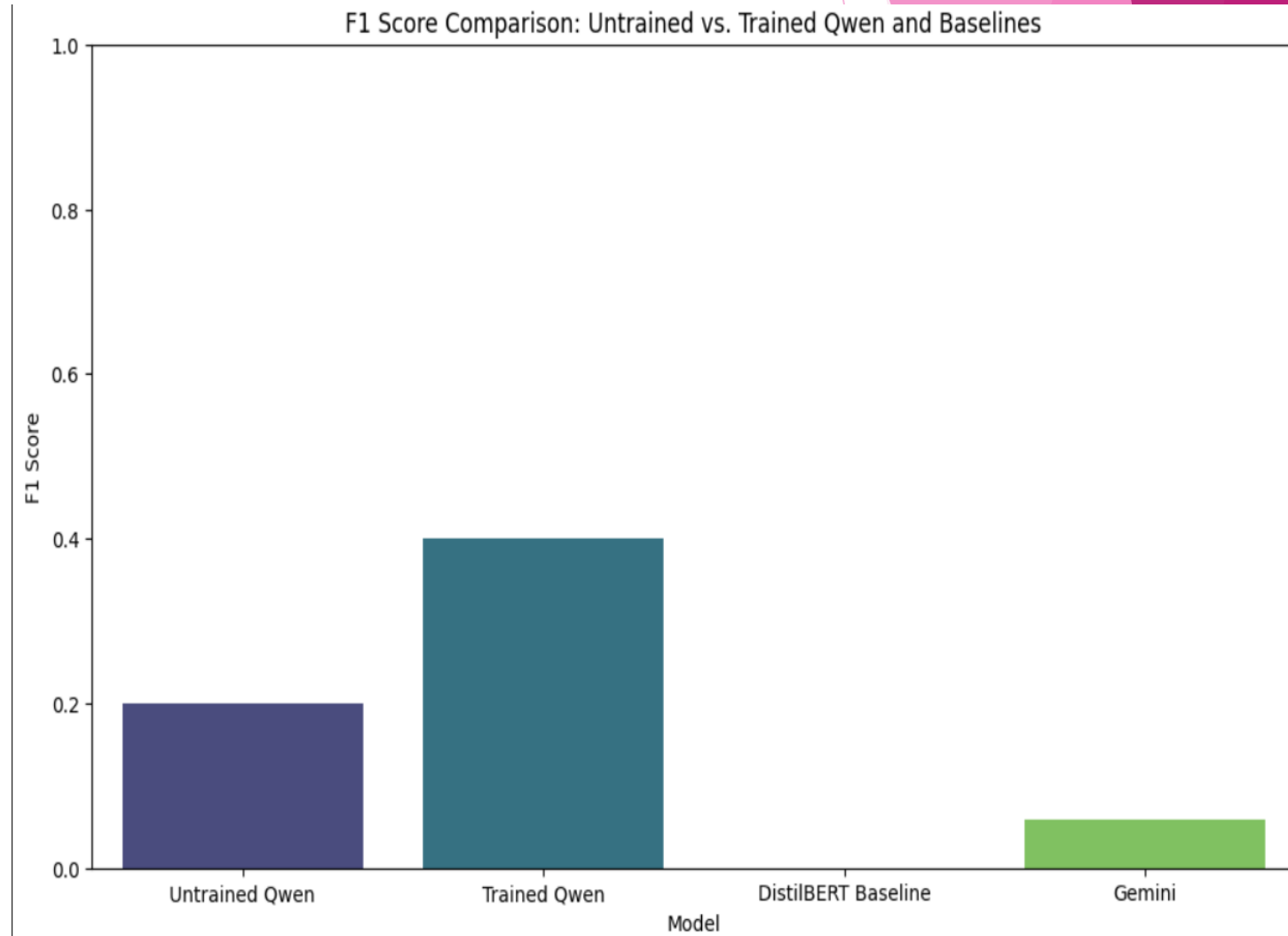
- ▶ Question Answering:
- ▶ 1.F1-score= 0.400
- ▶ 2.Exact Match= 0.000
- ▶ 3.relevance= 0.575

Metrics for Untrained Model:

F1 Score: 0.400

Exact Match: 0.000

Relevance Score: 0.575



Model Comparison:

All Model Comparison Results:

	Language	Model	Prediction	Gold Answer	Latency	F1	EM	Relevance
0	English	Untrained Qwen (SQuAD)	France	France	4.794604	1.000000	1.0	1.000000
1	English	Trained Qwen (SQuAD)	Franceарат奴隶应对理解ulanulanаратаратолодаратарата...	France	4.906376	0.016000	0.0	0.257305
2	English	DistilBERT (SQuAD)	France	France	0.262989	1.000000	1.0	1.000000
3	English	Gemini (SQuAD)	ERROR: 401 Client Error: Unauthorized for url:...	France	NaN	0.000000	0.0	0.052326
4	English	Llama (SQuAD)	egn\n\nNormandy is located on the northwestern...	France	3.665426	0.048780	0.0	0.346618
5	Chinese	Untrained Qwen (CMRC)	光荣与w-force合作开发的。	光荣和w-force	3.866897	0.615385	0.0	0.821322
6	Chinese	Trained Qwen (CMRC)	光荣аратalyzeарат理解ulan奴隶аратulle奴隶аратолодulle...	光荣和w-force	4.976809	0.016667	0.0	0.328061
7	Chinese	DistilBERT (CMRC)	2 光荣和w-force	光荣和w-force	0.847305	0.000000	0.0	0.063543
8	Chinese	Gemini (CMRC)	ERROR: 401 Client Error: Unauthorized for url:...	光荣和w-force	NaN	0.058824	0.0	-0.000464
9	Chinese	Llama (CMRC)	[INST] <<SYS>>\nYou are a helpful, respectful ...	光荣和w-force	3.728289	0.032154	0.0	0.190840

English vs Chinese Results:

Results: English (SQUAD) vs Chinese (CMRC)

